

Report on YOLOv26 Dataset Configuration (Meta Files)

1. Introduction

YOLO (You Only Look Once) models, including YOLOv26, rely on structured dataset configuration files (meta files) to train object detection systems. These files define dataset paths, class labels, and annotation formats, ensuring the model correctly interprets training data.

2. Dataset YAML Configuration File

The .yaml file is the main configuration file. It provides:

Key Components:

- **Dataset Paths**
 - train: → path to training images
 - val: → path to validation images
 - test: (optional) → path to test images
- **Number of Classes**
 - nc: → total number of object classes
- **Class Names**
 - names: → list of class labels

Example:

```
train: /data/images/train
val: /data/images/val
```

```
nc: 3
names: ["drone", "bird", "plane"]
```

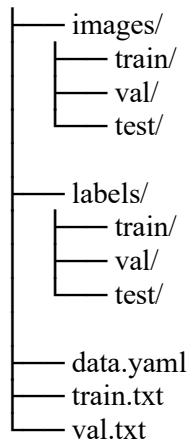
What it reveals:

- Structure of dataset directories
 - Object categories the model will learn
 - Separation of training and validation data
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3. Directory Structure

A typical YOLO dataset follows this structure:

```
dataset/
|
```



Insights:

- Each image has a corresponding label file
 - Train/val/test splits improve model generalization
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4. Label Files (.txt Format)

Each image has a label file with the same name.

Format:

<class_id> <x_center> <y_center> <width> <height>

Example:

0 0.45 0.52 0.30 0.20

Explanation:

- class_id → index from YAML class list
- Coordinates are normalized (0 to 1)
- Values represent bounding box position

What it reveals:

- Object location inside image
 - Object category
 - Model learns spatial relationships
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5. train.txt and val.txt Files

These files list image paths.

Example:

```
/data/images/train/img1.jpg  
/data/images/train/img2.jpg
```

Purpose:

- Helps model locate images during training
 - Useful when dataset is large or stored across locations
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6. Importance of Meta Files

Meta files are critical because they:

- Define dataset structure
- Ensure correct mapping between images and labels
- Enable reproducibility of experiments
- Allow easy dataset scaling and modification

Poorly structured meta files can lead to incorrect training and poor model performance.

7. Tools for Dataset Creation

Popular tools for creating YOLO datasets:

- Label Studio
- LabelImg
- Roboflow

These tools generate annotation files, dataset splits, and YAML configuration files.

8. References

- Ultralytics YOLO Documentation: <https://docs.ultralytics.com>

- Label Studio Documentation: <https://labelstud.io/guide/>
 - YOLO Dataset Format: <https://docs.ultralytics.com/datasets/detect/>
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9. Conclusion

YOLOv26 dataset meta files (YAML, labels, and text lists) provide a structured way to define object detection datasets. Understanding these files is essential for building accurate and efficient models. Proper dataset organization and high-quality annotations directly impact model performance.